

MA385 Part 1: Solving nonlinear equations

1.6: Fixed Point Iteration

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Newton's method can be considered to be a special case of a very general approach called *Fixed Point Iteration* (**FPI**) or *Simple Iteration*.

The basic idea is:

*If we want to solve $f(x) = 0$ in $[a, b]$, find a function $g(x)$ such that, if τ is such that $f(\tau) = 0$, then $g(\tau) = \tau$.
Choose x_0 and set $x_{k+1} = g(x_k)$ for $k = 0, 1, 2, \dots$*

0. Outline

1 Introduction

2 How not to choose g

3 Fixed points and contractions

■ Fixed Point

4 How many iterations?

5 Newton's method as a FPI

6 Exercises

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For more details, see Section 1.4 (Relaxation and Newton's method) of Süli and Mayers, *An Introduction to Numerical Analysis*

Also, Chapter 3 of Epperson:

https://search.library.nuigalway.ie/permalink/f/3b1kce/TN_cdi_askewsholts_vlebooks_9781118730966

1. Introduction

Yet again, we want to solve

Given a function $f(x)$, find $\tau \in [a, b]$ such that $f(\tau) = 0$.

Again, we'll try to find a sequence $\{x_0, x_1, \dots, x_k, \dots\}$, such that $x_k \rightarrow \tau$ as $k \rightarrow \infty$.

In this section, we'll consider *one-step* methods, which, like Newton's method, compute x_{k+1} just from x_k .

The Method is called **Fixed Point Iteration** (FPI):

- ▶ Choose a function g such that, if $f(\tau) = 0$, then τ is a fixed point of g .
- ▶ Choose x_0 , and then iterate with $x_{k+1} = g(x_k)$.

Note: Bisection & Secant are two-step methods.

1. Introduction

Example 1.6.1

Suppose $f(x) = e^x - 2x - 1$ and we are trying to find a solution to $f(x) = 0$ in $[1, 2]$. Then we can take $g(x) = \ln(2x + 1)$.

If we take $x_0 = 1$, then we get the following sequence:

k	x_k	$ f(x_k) $	$ \tau - x_k $
0	1.0000	0.2817	2.5643e-01
1	1.0986	0.1972	1.5782e-01
2	1.1623	0.1273	9.4148e-02
3	1.2013	0.0781	5.5092e-02
4	1.2246	0.0464	3.1868e-02
5	1.2381	0.0271	1.8310e-02
6	1.2460	0.0157	1.0479e-02
$\vdots$	$\vdots$	$\vdots$	
10	1.2553	0.0017	1.1079e-03

2. How not to choose g

We have to be quite careful with this method: **not every choice is g is suitable.**

For example, suppose we want the solution to $f(x) = x^2 - 2 = 0$ in $[1, 2]$. We could choose $g(x) = \underbrace{x^2 + x - 2}$. Then, if take $x_0 = 1$ we get the sequence:

$$x_0 = 1 \quad x_1 = g(x_0) = 1^2 + 1 - 2 = 0$$

$$x_1 = 0 \quad x_2 = g(x_1) = -2$$

$$x_2 = -2 \quad x_3 = g(x_2) = (-2)^2 + (-2) - 2 = 0$$

$$x_3 = 0 \quad x_4 = -2 \quad x_5 = 0 \quad x_6 = -2 \dots$$

2. How not to choose g

Before we do that in a formal way, consider the following...

Example 1.6.2

Use the Mean Value Theorem to show that the fixed point method $x_{k+1} = g(x_k)$ converges if $|g'(x)| < 1$ for all x near the fixed point.

By the MVT, there is a point $c \in [a, b]$ such that
$$\frac{g(b) - g(a)}{b - a} = g'(c).$$

$$\Rightarrow g(b) - g(a) = g'(c)(b - a).$$

Take $a = x_k$ & $b = \tau$ so

$$\underbrace{g(\tau)}_{\tau} - \underbrace{g(x_k)}_{x_{k+1}} = g'(c)(\tau - x_k) \quad \text{for some } c \in [x_k, \tau]$$
$$\Rightarrow |\tau - x_{k+1}| < |\tau - x_k|.$$

2. How not to choose g

This previous example is useful in two ways:

1. It introduces the tricks of using that $g(\tau) = \tau$ & $g(x_k) = x_{k+1}$.
2. It leads us towards the **contraction mapping theorem**.

Definition: fixed point

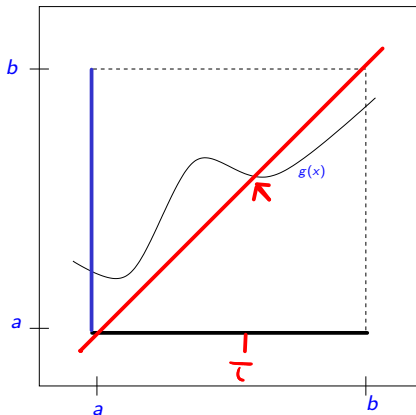
A function $g : \mathbb{R} \rightarrow \mathbb{R}$ is said to have a **fixed** point at $x = \tau$ if $g(\tau) = \tau$.

ϵ_g $g(x) = x + 1$ does not have a fixed point.

ϵ_g If g has a fixed point,
 $\&$ $f(x) = g(x) - x$ then
 $f(x) = 0$ has a solution.

Theorem 1.6.1 (Fixed Point Theorem)

Suppose the function g is continuous on $[a, b]$, and $a \leq g(x) \leq b$ for all $x \in [a, b]$. Then $g(x)$ has a *fixed point* in $[a, b]$.



Proof: Let $h(x) = g(x) - x$.

Then $h(a) = g(a) - a \geq 0$

Since $g(a) \geq a$

Also $h(b) = g(b) - b \leq 0$

since $g(b) \leq b$.

So, By the IVT,

$\exists \tau \in [a, b]$ s.t. $h(\tau) = 0$

Definition: **Contraction**

Let g be a continuous function defined on $[a, b]$. If there is a number $L \in (0, 1)$ such that

$$|g(\alpha) - g(\beta)| \leq L|\alpha - \beta| \text{ for all } \alpha, \beta \in [a, b], \quad (1)$$

then we say that g is a **contraction** on $[a, b]$.

Note : $L \in (0, 1) \Rightarrow 0 < L < 1$
In particular $L < 1$

Theorem 1.6.2 (Contraction Mapping Theorem)

Suppose that the function g is a real-valued, defined, continuous, and

- (a) maps every point in $[a, b]$ to some point in $[a, b]$, and
- (b) is a contraction on $[a, b]$,

then

- (i) $g(x)$ has a fixed point $\tau \in [a, b]$,
- (ii) the fixed point is unique,
- (iii) the sequence $\{x_k\}_{k=0}^{\infty}$ defined by $x_0 \in [a, b]$ and $x_k = g(x_{k-1})$ for $k = 1, 2, \dots$ converges at least linearly to τ .

(i) This is already proved as Thm 1.6.1.

(ii) To prove that the fixed point is unique, ... Suppose there are two fixed points τ_1 and τ_2 , and $\tau_1 \neq \tau_2$

$$\begin{aligned} \text{Then } |\tau_1 - \tau_2| &= |g(\tau_1) - g(\tau_2)| \quad [\text{fixed pts}] \\ &\leq L |\tau_1 - \tau_2| \quad [\text{contraction}] \\ &\quad [L \in (0, 1)] \end{aligned}$$

So $|\tau_1 - \tau_2| < |\tau_1 - \tau_2|$
which is not possible. So g has a unique fixed point.

3. Fixed points and contractions

Fixed Point

(iii) To prove that $\lim_{k \rightarrow \infty} x_k = \tau$,

Use that

$$|\tau - x_{k+1}| = |g(\tau) - g(x_k)| \leq L |\tau - x_k|$$

$$\text{In particular } |\tau - x_1| \leq L |\tau - x_0|$$

$$\text{And } |\tau - x_2| \leq L |\tau - x_1| \leq L^2 |\tau - x_0|$$

$$\text{In general } |\tau - x_k| \leq L^k |\tau - x_0|.$$

Then, since $0 < L < 1$.

$$\lim_{k \rightarrow \infty} L^k = 0. \quad \text{So } \lim_{k \rightarrow \infty} |\tau - x_k| = 0.$$

Note: this also gives

$$\lim_{k \rightarrow \infty} \frac{|\tau - x_{k+1}|}{|\tau - x_k|} = L \Rightarrow \text{At least linear convergence.}$$

4. How many iterations?

The algorithm generates a sequence $\{x_0, x_1, \dots, x_k\}$. Eventually we must stop. Suppose we want the solution to be accurate to say 10^{-6} , how many steps are needed? That is, how big do we need to take k so that

$$|x_k - \tau| \leq 10^{-6}?$$

The answer is obtained by first showing that

$$|\tau - x_k| \leq \frac{L^k}{1-L} |x_1 - x_0|. \quad (2)$$

This is because $|\tau - x_0| = |\tau - x_1 + x_1 - x_0|$
 $\leq |\tau - x_1| + |x_1 - x_0|$

$$\Rightarrow |\tau - x_0| \leq |g(\tau) - g(x_0)| + |x_1 - x_0|$$
$$\leq L |\tau - x_0| + |x_1 - x_0|$$

$$\text{so } |\tau - x_0| \leq \frac{1}{L-1} |x_1 - x_0|.$$

4. How many iterations?

Example 1.6.3

Suppose we are using FPI to find the fixed point $\tau \in [1, 2]$ of $g(x) = \ln(2x+1)$ with $x_0 = 1$, and we want $|x_k - \tau| \leq 10^{-6}$, then we can use (2) to determine the number of iterations required.

Here $g'(x) = \frac{2}{2x+1}$. Since $x \in [1, 2]$

$|g'(x)| \leq \frac{2}{3}$ for all $x \in [1, 2]$.

So take $L = 2/3$.

(check!) can show
that $\textcircled{37}$ iterations are needed.

5. Newton's method as a FPI

Newton's method can be thought of as an example of a fixed point method, where we take

$$g(x) = x - \frac{f(x)}{f'(x)}.$$

However, we know that, when Newton's Method converges it does so quadratically, whereas FPI converges (at least linearly).

5. Newton's method as a FPI

Let's remind ourselves of the definition:

- ▶ We have a sequence of numbers $\varepsilon_0, \varepsilon_1, \dots$, such that $\lim_{k \rightarrow \infty} \varepsilon_k = 0$.
- ▶ These bound the errors: $|\tau - x_k| \leq \varepsilon_k$ Let $\tau = \lim_{k \rightarrow \infty} x_k$.
- ▶ We know that $\lim_{k \rightarrow \infty} x_k = \tau$
- ▶ Then we say that the sequence $\{x_k\}_{k=0}^{\infty}$ converges *with at least order q* if

$$\lim_{k \rightarrow \infty} \frac{\varepsilon_{k+1}}{(\varepsilon_k)^q} = \mu,$$

for some constant μ .

For $q = 1$ we get linear convergence. If $q = 2$, we say it is *quadratic*.

and $\mu \in (0, 1)$

5. Newton's method as a FPI

Suppose that we have a convergent Fixed Point Method,
 $x_{k+1} = g(x_k)$, but with the additional property that $g'(\tau) = 0$.
Then, in fact, FPI converges (at least) quadratically):

Use a Truncated Taylor Series :

$$g(x) = g(\tau) + (x-\tau)g'(\tau) + \frac{1}{2}(x-\tau)^2 g''(\eta)$$

$\stackrel{!}{=} \tau$ for some $\eta \in (x, \tau)$

Take $x = x_k$ & use $g(\tau) = \tau$ & $g'(\tau) = 0$ to get
 $g(x_k) = \tau + \frac{1}{2}(x_k - \tau)^2 g''(\eta)$. Now use $g(x_k) = x_{k+1}$.

$$\text{So } x_{k+1} - \tau = \frac{1}{2}(x_k - \tau)^2 g''(\eta)$$

$$\text{Thus } \frac{|x_k - \tau|}{|x_{k+1} - \tau|} = \frac{1}{2} g''(\eta). \quad \text{Since } \eta \in (x_k, \tau), \text{ and}$$

as $k \rightarrow \infty$, so $x_k \rightarrow \tau$. Thus $\eta \rightarrow \tau$.

5. Newton's method as a FPI

Suppose that we have a convergent Fixed Point Method, $x_{k+1} = g(x_k)$, but with the additional property that $g'(\tau) = 0$. Then, in fact, FPI converges (at least) quadratically):

Therefore
$$\lim_{k \rightarrow \infty} \frac{|\tau - x_{k+1}|}{|\tau - x_k|^2} = \mu$$

where μ is the constant $\frac{1}{2} g''(\tau)$.

5. Newton's method as a FPI

Equipped with this, one can show that, in the FPI setting, Newton converges quadratically. See Exercise.

1. Set $g(x) = x - \frac{f(x)}{f'(x)}$.
2. Find $g'(x)$.
3. Show $g'(\tau) = 0$

6. Exercises

Exercise 1.6.1

Is it possible for g to be a contraction on $[a, b]$ but not have a fixed point in $[a, b]$? Give an example to support your answer.

Exercise 1.6.2

Show that $g(x) = \ln(2x + 1)$ is a contraction on $[1, 2]$. Give an estimate for L . (Hint: Use the Mean Value Theorem).

6. Exercises

Exercise 1.6.3

Suppose we wish to numerically estimate the famous *golden ratio*, $\tau = (1 + \sqrt{5})/2$, which is the positive solution to $x^2 - x - 1$. We could attempt to do this by applying fixed point iteration to the functions $g_1(x) = x^2 - 1$ or $g_2(x) = 1 + 1/x$ on the region $[3/2, 2]$.

- (i) Show that g_1 is *not* a contraction on $[3/2, 2]$.
- (ii) Show that g_2 is a contraction on $[3/2, 2]$, and give an upper bound for L .

6. Exercises

Exercise 1.6.4

In class we saw that if $g(\tau) = \tau$, and the fixed point method given by

$$x_{k+1} = g(x_k),$$

converges to the point τ (where $g(\tau) = \tau$), and if $g'(\tau) = 0$, then the method converges quadratically.

Show that, in fact if

$$g'(\tau) = g''(\tau) = \cdots = g^{(p-1)}(\tau) = 0,$$

then it converges with order p .

Exercise 1.6.5 (Based on Q2(d), Semester 1 exam, 25/26)

Show that Newton's method can be written as a fixed-point method.

Use the result on Slide 17 to show it converges quadratically.